Diagnosis of SMGF in ESOP based Reversible Logic Circuit

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Abstract—**Testing of reversible circuit is an important issue to assure the quality and reliability of the reversible quantum circuit. Several fault models like Single Missing Gate Fault (SMGF), Partial Missing Gate Fault (PMGF), Multiple Missing Gate Fault (MMGF) and Repeated Gate Fault (RGF) have been predicated for better representation of faults in reversible quantum circuits. In this work, we develop an algorithm for diagnosis of Single Missing Gate Fault (SMGF) in exclusive-or sum of product (ESOP) based reversible circuit. Without using conventional testing technique where test vectors are generated to detect the fault, we present boolean expression based testing technique which produces the faulty gate details as a product term in boolean expression. The developed testing technique first detects the fault, then identifies the fault and finally corrects the erroneous part of the circuit to make it fault free.**

Keywords: - Reversible logic; PPRM; ESOP; diagnosis; missing gate fault; testing

I. Introduction

The traditional irreversible logic computation produces certain amount of heat due to the loss of each bit of information regardless of underlying technology [1]. As the reversible computing does not lose any information [1], reversible logic can be used to design information lossless circuits, and therefore, it has the potential to reduce the power dissipation drastically [1, 2]. Reversible logic appears to be promising due to its wide application in the emerging technologies. Some of the emerging technologies where the reversible logic are used, are quantum computing, quantum dot cellular automata, optical computing, spintronics, molecular computing, cryptography, nanotechnology, DNA technology, and low-power CMOS design [3, 4, 5].

Since quantum gates or circuits satisfy 'no-cloning' behaviour, and are information lossless, they do not permit fan-out, and have an equal number of inputs and outputs. A reversible combinational circuit should consist of only reversible gates, which implement reversible functions; such gates need to be specially designed, e.g., by using controlled-not (CNOT), Feynman [6], Fredkin [7] and Toffoli or Toffoli [8] gates. Recently, researchers are taking interest on the prediction of reversible fault models and development of the techniques for testing reversible circuits [10-15]. The universal testability of reversible logic circuits implemented with k-CNOT gates under the stuck-at fault model (both single and multiple) has been investigated in [13]. Several new fault models predicated for testing the reversible quantum circuit are single missing-gate fault (SMGF) [11], repeated-gate fault (RGF) [16], partial missing-gate fault (PMGF) [17], and multiple missing-gate faults (MMGF) [22]. Determination of a universal test set for the detection faults in the reversible circuits has been studied for these fault models [12, 13]. It has been shown earlier that in an ($n\times n$) reversible circuit implemented with k-CNOT gates, the addition of only one extra control line along with duplication of each k-CNOT gate yields an easily testable design, which admits a universal test set of size (n+1) that detects all SMGFs, RGFs, and PMGFs [16, 17]. Online detection of any single bit fault of an ESOP circuit [28] has been presented in [18]. Application of boolean satisfiability and pseudo-boolean optimization to generate automatic test patterns using complementary approaches has been introduced in [29]. Apart from these techniques, in [30] an improved approach for diagnosis of faults in the reversible circuits has been presented.

In this work, we develop an algorithm to detect and identify SMGF fault in an ESOP circuit. The proposed algorithm is implemented in two phases. First phase checks the possible faults in the testable realization, where Positive Polarity Reed-Muller (PPRM) form has been evaluated to perform exclusive 'or' operation with testable circuit. In second phase, the identification of fault and necessary correction in the faulty circuit are performed. The proposed testing technique is based on boolean expression.

The rest of the paper is organized as follows. A general discussion on the reversible logic and reversible gates is presented in Section II. In Section III, we present the proposed technique to diagnosis the fault in ESOP based circuit. The discussion is concluded in Section IV.

II. Background

A. *Reversible Logic Circuit:*

A fan-out free circuit (C_{nf}) with circuit depth (d) over the set of input lines $X=\{x_1, x_2, \ldots, x_n\}$ is said to be reversible (R_c) [9, 20] if the circuit consists of the reversible gates (g_i) [9] only and the number of inputs (m) is equal the number of outputs (n) i.e. $C_{nf}= g_0 . g_1 . g_2 . \ldots . g_{(d-1)}$ where g_i represents i^{th} reversible gate of the circuit.

An irreversible function can be converted into the reversible function by adding extra outputs, called *garbage* to make the number of output lines equal to the number of input lines. If the maximum number of identical output vector is p, then the total number of garbage outputs needed is equal to $[\log_2 p]$ [19]. Most of the traditional logic gates such as AND, OR, NAND, NOR, and EXOR are not reversible. The only conventional NOT gate is reversible gate.

B. Reversible gates

Definition 1: A reversible gate is defined as $g(C; T)$ where C is called the set of control lines and T is the set of target lines

978-1-4799-6965-4/14 $31.00 © 2014 IEEE
DOI 10.1109/ISED.2014.26

with $C \cap T = \Phi$. In the reversible gate, the control set C may be empty but T contains at least one target line.

Definition 2: For the domain variables $\{x_1, x_2, \dots, x_n\}$, the multiple control Toffoli gate has the form $TOF(C;t)$, where $C = \{x_{i_1}, x_{i_2}, \dots, x_{i_k}\}$, $t = \{x_j\}$ and assures $C \cap t = \phi$. It maps the input vector $(x_1^0, x_2^0, \dots, x_n^0)$ to boolean pattern $\{x_1^0, x_2^0, \dots, x_{j-1}^0, x_j^0 \oplus x_{i_1}^0 x_{i_2}^0 \dots x_{i_k}^0, x_{j+1}^0, \dots, x_n^0\}$. The set C that controls the change of jth bit is the set of controls and t is called the target.

C. *Reed-Muller form*

Boolean algebra operators play a crucial role in logic synthesis and testing of digital circuits. The simplest case of the Galois field algebra is modulo-2 algebra and the representation of any boolean function is possible by modulo-2 algebra. In the practical field of applications, Reed-Muller expansion is nothing but modulo-2 sum-of product expression. The construction of the reversible logic circuit by modulo-2 operations can be realized by means of exclusive OR (EXOR) gates. Any boolean function $f(x_1 x_2 \dots x_n)$ of n variables can be represented by the following forms of the Reed-Muller expansion [26, 27].

Positive polarity Reed-Muller (PPRM): The PPRM form is an EXOR sum of products based canonical expression, where each variable has positive polarity (not complemented form). An n variable PPRM expression can be represented as

$f(x_1 x_2 \dots x_n)=$

$a_0 \oplus a_1 x_1 \oplus a_2 x_2 \oplus a_3 x_1 x_2 \oplus \dots \oplus a_{2^n-1} x_1 x_2 \dots x_n$,

where $a_i \in \{0, 1\}$.

The variable a_i represents the coefficient vector, where x_i are input variable terms. If the coefficient is 0, then the associated product term does not appear in the PPRM expression. If the coefficient is 1, then the associated product term appears in the PPRM expression.

Fixed polarity Reed-Muller (FPRM): In FPRM form, each literal appear as complemented or un-complemented form throughout the expression i.e. polarity of each variable is fixed. For an n variable Boolean function there exists 2^n number of different FPRM expression. An n variable FPRM expression can be represented as

$f(x_1 x_2 \dots x_n)=a_0 \oplus a_1 \dot{x}_1 \oplus a_2 \dot{x}_2 \oplus a_3 \dot{x}_1 \dot{x}_2 \oplus \dots \oplus a_{2^n-1} \dot{x}_1 \dot{x}_2 \dots \dot{x}_n$

,where $a_i \in \{0,1\}$ and $\dot{x} \in \{x, \bar{x}\}$.

Mixed polarity Reed-Muller: The MPRM expression is the generalized form of FPRM expression where there is no restriction in the polarity of each variable. For an n variable boolean function, there exists 3^n number of MPRM expression. An n variable MPRM expression can be represented as

$f(x_1 x_2 \dots x_n) = a_0 \oplus a_1 \dot{x}_1 \oplus a_2 \dot{x}_2 \oplus a_3 \dot{x}_1 \dot{x}_2 \oplus \dots \oplus a_{2^n-1} \dot{x}_1 \dot{x}_2 \dots \dot{x}_n$

,where $a_i \in \{0,1\}$ and $\dot{x} \in \{x, \bar{x}\}$.

C. *ESOP based circuit*

An ESOP [25] form is a variation of the *Sum-Of-Products* (SOP) form in boolean function representation. Any SOP-based boolean function can be converted to the equivalent ESOP form. For example, $f = xy + yz$ is in SOP form. Now, if we convert it to ESOP form, then the equivalent function becomes $f = xy \oplus yz \oplus xyz$. But, a definite mapping algorithm is required to convert a SOP form to equivalent ESOP form.

To construct an ESOP-based circuit corresponding to input function, it is important to define gate specifications for design of the circuit. Detailed specifications of gate and its control structure are kept in a separated list termed as *cube list*. A cube list is represented by a matrix (M) of size ($2^n \times n$), where each row represents a cube (C_i). The cube list generates the function $f(x_1, x_2, \dots, x_n)$, which is a collection of product terms separated by exclusive-or ($\oplus$) operators. Each product term is represented by a cube (C_i), which is a vector in a vector space of dimension 2^n. An ESOP cube C_i is represented as $C_i = \langle a_1\ a_2\ a_3 \dots a_n \rangle$, where a_i ∈ {0, 1, −} for function $f(x_1, x_2, \dots, x_n) = C_1 \oplus C_2 \oplus \dots \oplus C_k$.

Each cube C_i is always mapped to a Toffoli gate T_i, where a defined bit value (0 or 1) of respective bit position in cube C_i corresponds to a control node in the Toffoli gate T_i. If the bit value of the kth position in cube C_i is "zero", then the control value of the kth position in Toffoli gate T_i is a negative control or if the bit value is 1, then there will be a positive control.

The ESOP-based circuit representation of an n bit single output function requires a total of $(n + 1)$ circuit lines, where the first n lines are termed as control lines and the last $(n + 1)^{th}$ line is the target line. The target line plays the dominant role in the entire circuit by providing the functional output value.

An example of an ESOP cube list and its equivalent ESOP representation are presented in Fig. 1(a) and in Fig. 1(b), respectively.

x_1	x_2	x_3	x_4
–	1	1	–
–	–	–	1
1	–	–	–
0	1	–	1
0	1	1	0

Fig. 1(a): Cube list for $f(x_1, x_2, x_3, x_4) = x_2 x_3 \oplus x_4 \oplus x_1 \oplus \bar{x}_1 x_2 x_4 \oplus \bar{x}_1 x_2 x_3 \bar{x}_4$.

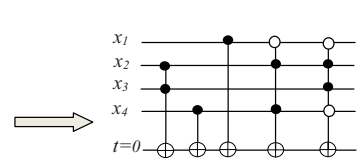


Fig. 1(b): Equivalent ESOP representation

E. *Fault Model*

A fault is a physical defect in a system. In other word, a system has a fault if there is a physical difference between the correct system and the current system [20]. A fault model simplifies the analysis and complexity of testing by reducing the number of faults in a circuit. Several new fault models for k-CNOT based reversible circuits have been introduced in [12, 15]. These are single missing-gate fault (SMGF) [22], the repeated gate fault (RGF) [22], partial missing-gate fault (PMGF) [22], and multiple missing-gate fault (MMGF) [22]. In this section, we briefly discuss the concept of different fault models.

Single missing-gate fault (SMGF): The SMGF model corresponds to the case when one gate completely disappears from the circuit.

Fig. 2(a) shows an SMGF marked by the dotted box in the reversible circuit ham3 design1 benchmark, where the first 2-CNOT gate is missing. An SMGF is detected by setting logic 1 value on all the control inputs of the gate, and any value either 0 or 1 on the target input as well as on the wires not

connected to the gate. In Fig. 2(a), if we apply (a, b, c) = (0, 1, 1) at the input of the circuit, the normal output would be (e, f, g) = (1, 0, 0), whereas, in the presence of the SMGF fault marked by the dotted box, the output will be (e, f, g) = (0, 1, 1). The number of possible SMGFs is equal to the number of gates in the circuit.

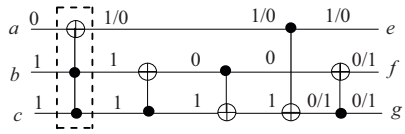


Fig. 2(a): Single Missing Gate Fault (SMGF)

Repeated-gate fault (RGF): A repeated gate fault (RGF) means an unwanted replacement of a gate by several instances of the same gate. An RGF may be used to model the occurrence of long or duplicated pulses.

Fig. 2(b) depicts that the first gate is repeated in the circuit ham3\design#1. The effect of this fault is same as that of an SMGF at the first 2-CNOT gate in the original circuit. If we apply {a, b, c} = {0, 1, 1}, the normal output would be {e, f, g} = {1, 0, 0}, whereas, in the presence of the above RGF marked by the dotted box, the output will be {e, f, g} = {0, 1, 1}. Hence, it is detected by the vector {a, b, c} = {0, 1, 1}. It is clear that if a RGF replaces a gate by even number of instances of the same gate, its effect is similar to the effect of the SMGF with respect to the same gate. If the RGF replaces a gate by odd number of instances of the same gate, the fault is redundant, i.e., it does not change the function of the circuit. Further, it has been shown that any SMGF test set detects all detectable RGFs.

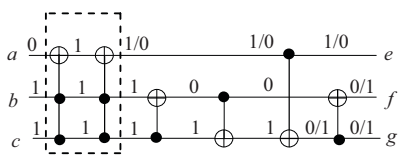


Fig. 2(b): RGF in ham3\design#1 reversible circuit

Partial missing-gate fault (PMGF): This is used to model the defects resulting from the partially misaligned or mistuned gate pulses. It changes a *k*-CNOT gate into a p-CNOT gate, with $p < k$. The corresponding fault is called as $(k\text{-}p)^{th}$ order PMGF.

Fig. 2(c) shows a first-order PMGF affecting the second control input of the leftmost gate. An SMGF can be seen as a 0-order PMGF. In Fig. 2(c), if we apply {a, b, c} = {1, 0, 1}, the normal output would be {e, f, g} = {1, 1, 0}, whereas, in the presence of the first order PMGF fault, the output will be {e, f, g} = {0, 0, 1}.

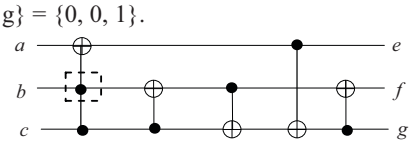


Fig. 2(c): PMGF in ham3\design#1 reversible circuit

Hence, the vector {a, b, c} = {1, 0, 1} detects this fault. It is true that a PMGF (of first or higher order) is detected when a '0' is applied to at least one of the affected control inputs and a '1' to all other control inputs. Thus, a higher order PMGF is detected by a test vector for a first order PMGF.

Multiple missing-gate faults (MMGF): This is defined as complete disappearance of two or more consecutive k-CNOT gates from the circuit. Fig. 2(d) shows that the circuit has an MMGF marked by the dotted box. This fault is detected by the vector {a, b, c} = {0, 1, 1}.

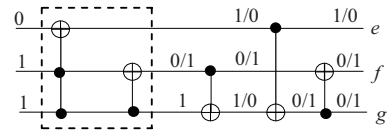


Fig. 2(d): MMGF in ham3ndesign#1

III. PROPOSED TECHNIQUE

In this work, we propose a method to diagnosis the single missing gate fault (SMGF) in an ESOP based reversible circuit. Diagnosis of SMGF in the testable circuit C_{test} can be performed in the following way. First, we evaluate the fault free ESOP circuit (C_{true}) from function specification table (T_{spec}). Now we perform logical XOR between C_{true} and C_{test}. But a problem appears while designing the fault free ESOP circuit C_{true} for function *f(n)* as there can be more than 3^n numbers of distinct ESOP circuit (as MPRM is a subclass of ESOP expression). Now to identify the fault free ESOP circuit (C_{true}) from more than 3^n numbers of distinct ESOP circuit needs some tedious computation. To avoid this situation, we choose such a form of function *f(n)* that has only one circuit in its class i.e. PPRM class.

There are two phases in our proposed testing technique. In initial phase, we test a reversible circuit whether it is faulty or fault-free. In second phase, we identify the fault and correct the circuit to compensate the faulty behaviour of the circuit with some gates such that the circuit becomes fault free.

A. *Detection of SMGF*

Let us consider a specification table (T_{spec}) for the circuit function (*f(n)*) and its equivalent testable ESOP circuit (C_{test}). To check the presence of SMGF in testable ESOP circuit (C_{test}), we need to pass all the possible 2^n bit pattern through circuit C_{test}. If a bit pattern that does not satisfy the input specification table (T_{spec}) is found, then C_{test} can be declared as faulty circuit. But this approach has large computational complexity (in best case O(*n*) and in worst case O(2^n)). To avoid this drawback, we use a simple approach in our proposed algorithm to detect the SMGF in an ESOP circuit.

We evaluate the PPRM based ESOP expression f_{true}^{PPRM} from the function specification table T_{spec} corresponding to input testable realization C_{test}. Next, we obtain the corresponding MPRM expression f_{test}^{MPRM} from testable ESOP circuit C_{test}. Now we change the polarity of each variable that is present in MPRM expression f_{test}^{MPRM} to positive one and obtain an equivalent FPRM logic expression (f_{test}^{FPRM}). We perform logical XOR operation between f_{true}^{PPRM} and f_{test}^{FPRM} i.e. $L_{XOR}=f_{true}^{PPRM} \oplus f_{test}^{FPRM}$. Here, we can decide whether the C_{test} is fault-free or faulty. If the computed value L_{XOR} is zero, that means it does not produce any product term and the testable

C_{test} is declared as fault-free, otherwise a SMGF is present in the C_{test}.

B. *Identification and correction of SMGF*

In detection phase, we decide whether a circuit is fault-free or not. Here, we identify the fault first and then correct the faulty behaviour of the circuit.

The output expression generated from L_{XOR} is the specification for missing gate details as a boolean expression and the variables of the boolean expression L_{XOR} are the control lines of the missing gate (M_g) of the faulty reversible circuit.

To correct the circuit, if missing gate (M_g) is appended to input testable circuit C_{test}, then circuit becomes fault free.

The technique for detecting and identifying single missing gate fault in ESOP circuit is presented in algorithm 1.

ALGORITHM 1: SMGF Detection and Identification of ESOP circuit

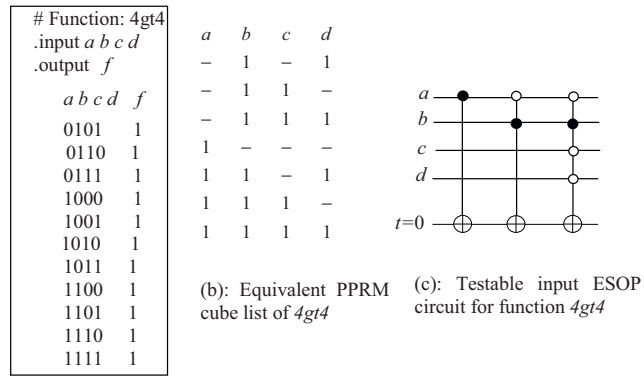
Input : Function (*f(n)*) specification file (T_{spec}) corresponding to testable input ESOP circuit (C_{test}).
Output: Control lines of the missing gate

begin
- Step1: Calculate PPRM form f_{true}^{PPRM} of *f(n)* from T_{spec} .
- Step2: Derive FPRM form f_{test}^{FPRM} of the testable reversible circuit C_{test}, where all the literals of FPRM expression are positive in polarity.
- Step3: Perform logical XOR operation between f_{true}^{PPRM} and f_{test}^{FPRM} to check whether there is any SMGF or not.
  Calculate $L_{XOR}=f_{true}^{PPRM}\oplus f_{test}^{FPRM}$
- Step4: IF (L_{XOR} ==∅) then
  - **begin**
    - 4.1 No SMGF fault is present in C_{test} .
  - ELSE
  - **begin**
    - 4.1 SMGF fault is present in C_{test} .
    - //Detection of SMGF//
    - 4.2 The variables of the boolean expression L_{XOR} that is generated from the above XOR operation are the control lines of the missing gate of the faulty reversible circuit.
    - //Identification of SMGF//

Here, we are showing the detection, identification and correction of SMGF in an ESOP circuit with examples.

Example 1: The benchmark circuit *f(n)=4gt4* [23] is taken as an input circuit to test the algorithm. Input specification of the file is shown in Fig. 3(a). The testable ESOP representation of the above benchmark file is depicted in Fig. 3(c). At first, we have to check whether it is fault-free or not. So, we extract the PPRM cover from the function specification table T_{spec} using the method presented in [24]. The obtained an equivalent PPRM cube list is shown in Fig. 3(b) and its corresponding PPRM expression is $f_{true}^{PPRM} = a \oplus bd \oplus bc \oplus bcd \oplus abd \oplus abc \oplus abcd$. Next, we find the MPRM expression corresponding to testable ESOP circuit C_{test} i.e. $f_{test}^{MPRM} = a \oplus \bar{a}b\,\bar{c}\,\bar{d} \oplus \bar{a}b$ and change the polarity of each variable in f_{test}^{MPRM} to positive one to obtain an equivalent FPRM expression ($f_{test}^{FPRM} = a \oplus bd \oplus bc \oplus bcd \oplus abd \oplus abc \oplus abcd$). Now, we perform logical XOR operation between f_{true}^{PPRM} and f_{test}^{FPRM} i.e. $L_{XOR} = f_{true}^{PPRM} \oplus f_{test}^{FPRM} = \emptyset$.

Here, we can confirm that there is no SMGF in the circuit C_{test} as L_{XOR} produces null terms.



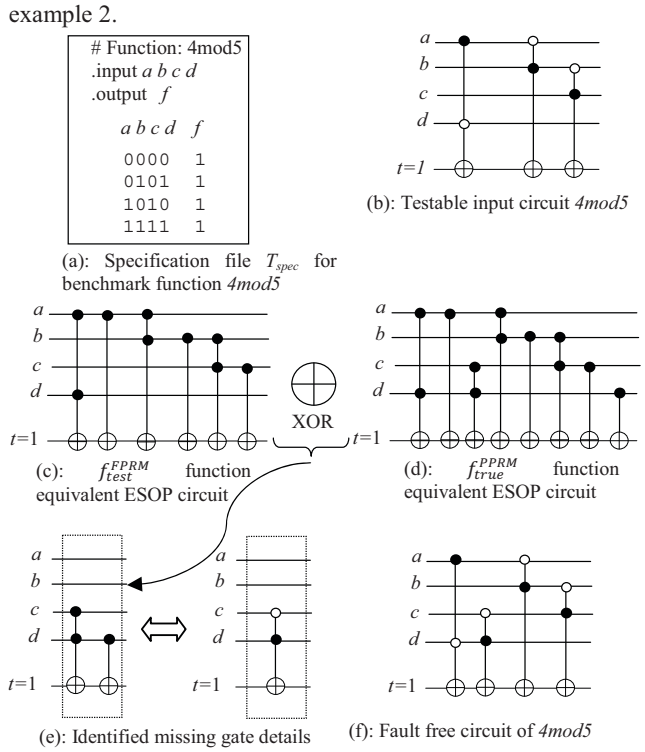
(a): Input specification file of *4gt4*

(b): Equivalent PPRM cube list of *4gt4*

(c): Testable input ESOP circuit for function *4gt4*

Fig. 3: Illustration of Example 1

Another example on the proposed detection method of single missing gate fault (SMGF) in an ESOP circuit is presented in example 2.



(a): Specification file T_{spec} for benchmark function *4mod5*

(b): Testable input circuit *4mod5*

(c): f_{test}^{FPRM} function equivalent ESOP circuit

(d): f_{true}^{PPRM} function equivalent ESOP circuit

(e): Identified missing gate details

(f): Fault free circuit of *4mod5*

Fig. 4: Illustration of Example 2

Example 2: Let us consider an ESOP circuit corresponding to input benchmark function *4mod5* [23]. We have to detect whether it is fault-free or not. If it has the fault, then we have to identify the fault and correct the circuit by appending some gate to compensate the faulty behavior.

Input specification table (T_{spec}) for testable ESOP function (*f(n)= 4mod5*) and its corresponding testable circuit (C_{test}) are

depicted in Fig. 4(a) and in Fig. 4(b), respectively. First, we evaluate the PPRM expression (f_{true}^{PPRM} = 1⊕ $ad \oplus ab \oplus bc \oplus cd \oplus a \oplus b \oplus c \oplus d$) from the function specification table T_{spec}. Next, we obtain the corresponding MPRM expression (f_{test}^{MPRM} =1⊕ $a\bar{d} \oplus \bar{a}b \oplus \bar{b}c$) from testable ESOP circuit C_{test}. Now, we change the polarity of each variable in f_{test}^{MPRM} to positive one and obtain an equivalent FPRM logic expression (f_{test}^{FPRM} =1⊕ $ad \oplus ab \oplus bc \oplus a \oplus b \oplus c$). Equivalent ESOP representations of f_{test}^{FPRM} and f_{true}^{PPRM} are presented in Fig. 4(c) and in Fig. 4(d), respectively. Now, we perform logical XOR operation between f_{true}^{PPRM} and f_{test}^{FPRM} i.e. $L_{XOR} = f_{true}^{PPRM} \oplus f_{test}^{FPRM} = cd \oplus d = \bar{c}d$.

Here, we can confirm that there is a SMGF fault in the circuit C_{test} as L_{XOR} produces product terms. The output expression generated from L_{XOR} is the specification for missing gate details as a boolean expression and the variables of the boolean expression L_{XOR} are the control lines of the missing gate M_g as shown in Fig. 4(e)) of the faulty reversible circuit. Now, we can say that the missing gate has a negative control at control line c and a positive control at control line d.

To correct the circuit, missing gate (M_g) is appended to input testable circuit C_{test} and the fault free ESOP representation for benchmark function *4mod5* is obtained as shown in Fig. *4(f)*.

IV. CONCLUSION

In this work, a boolean expression based SMGF testing technique in ESOP circuit has been presented. The designed algorithm successfully detects, identifies the fault and then corrects the erroneous part of the circuit by appending detected missing gate to the erroneous circuit to make it fault free. Several ESOP benchmark circuits have correctly been executed on the proposed algorithm, which establishes the novelty of our approach.

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